Spain
National Commission for Markets and Competition

I. Short summary

Last year, the Spanish National Markets and Competition Commission's (CNMC) contribution to the 2nd Annual Report Stanford Computational Antitrust addressed the topic of Computational tools for detecting collusion.

Taking into consideration that the CNMC has among its top priorities adapting its competition enforcement policy to the current wave of digitisation and use of artificial intelligence, the initiatives already in course at that moment have experienced an evolution and, in some respects, a relevant breakthrough.

As it was then reported, the CNMC has an Economic Intelligence Unit (EIU) devoted to, among other activities, applying new techniques of data analysis, using artificial intelligence to detect not only new forms of (namely, algorithmic) anticompetitive conduct, but also traditional ones, subtler and more refined every day. The unit also aims to increase *ex officio* detection and strengthen the robustness of other investigations.

The availability of each day a more ample database covering all aspects of public procurement, including detailed factual data of enterprises and related persons thanks to the agreement signed between CNMC and Real Estate Registers Official College of Spain, has been also an important branch of activity for the EIU, providing our data analysis tools a more solid base to yield relevant results.

The CNMC thus remains committed to progress in applying new techniques to ensure that competition is safeguarded in all markets, especially in public procurement. The scope of this contribution will be to summarily describe the actual use of some of them and their stage of development.

II. Detection and analysis of anticompetitive conduct in the Economic Intelligence Unit

Algorithms and other information technology tools can be used for anticompetitive purposes, by means of software tools with a varied degree of sophistication (pricing algorithms, etc.). But anticompetitive conduct can also appear in other forms, such as self-preferencing or discrimination (e.g. in rankings or matching algorithms). These risks appear more often in digital business models where algorithmic tools are used frequently, but they can also take place in other sectors.

To tackle these challenges, the Spanish National Markets and Competition Commission (CNMC) introduced digitisation as a priority in its Strategic Plan,[8] leading to different initiatives in its annual Action Plans.[9] One of such initiatives was the creation of an Economic Intelligence Unit (EIU), to take advantage of new digital techniques as an aid to detect different kinds of anticompetitive conduct, including when this conduct is facilitated by algorithms and other advanced tools.

This contribution summarizes and updates the work carried out by the Economic Intelligence Unit (EIU) to increase the detection of anticompetitive conduct through digital tools.

The analysis carried out by the EIU is used both for purely *ex officio* detection and for refining and completing the analysis in other cases: enhancing and reinforcing evidence, adding information to existing complaints, dawn-raid planification through OSINT information, etc.

Collecting the right data allows to use specific techniques and analyses, such as supervised machine-learning, graph analysis, and other advanced techniques related to data management and processing.

Prior to the creation of the EIU itself, the CNMC worked since 2015 on the construction of a public procurement database, developing specific algorithms for data extraction and cleaning, categorizing the data by quality levels regarding all participating bids and bidders.

Regarding public procurement, the CNMC database is the result of the aggregation of different sources. Firstly, the Public Sector Procurement Platform (PSPP) is a completely up-to-date electronic platform publishing all the calls for tenders and their outcomes. It encloses data from regional and local contracting bodies, since platforms of the different government authorities and public entities

[8] CNMC, *Plan Estrategico*, https://www.cnmc.es/sobre-la-cnmc/plan-estrategico (last accessed June 8, 2024).
[9] CNMC, *Plan de Actuacion*, https://www.cnmc.es/sobre-la-cnmc/plan-de-actuacion (last accessed June 8, 2024).

are interconnected to establish a single platform that centralises the publication of public sector procurement. Thereafter this repository is completed with data from the procurement bodies that are not integrated within PPSP, such as the centralized contracts and with the non-standardized documents associated with the tenders, such as notice of award or tender amendments.

In addition to it, the CNMC also downloads the Ministry of Finance Centralised Procurement Portal information of framework agreements, centralised contracts, and other centralised procedures for minor contracts. So far, the database contains more than 3.5 million contracts and near 4 million lots.

Therefore, this CNMC's public procurement database could be considered one of the most complete at national level, with structured data about tenders from all public administration levels and types, but also the related documents (non-structured data).

In addition to all these, in 2023 there arose another relevant source of data available due to the agreement signed between CNMC and Real Estate Registers Official College of Spain. Thanks to it, CNMC has access to all the official economic and financial information about the companies acting as bidders in public tenders, such as the location of business premises, main stockholders and/or controllers of the company, successor firms in case of extinction, etc.

With the combination of data coming from all these sources, the CNMC has hence created a complete digital database of public procurement in Spain that has allowed the development in recent times of several powerful tools that take advantage of the processed data and currently help EIU to detect potential cases of bid-rigging.

In the first place, the CNMC has a comprehensive set of analysis tools supporting detection. These are mainly business intelligence tools, like Microsoft Power BI for visualization and analysis, along with other in-house developed, dedicated web-based searching tools. These are in some aspects conventional tools, but tuned to deal efficiently with great amounts of data, giving our analysts the ability for discovering subtle trends in such data, or to relate evidence from a particular tender to other activities of the subjects under investigation.

In addition to it, we have taken advantage of graph-oriented databases as a source of screening variables, based on graph metrics calculation, to serve as inputs for the machine learning process (*see* below). Such databases store nodes and relationships, instead of tables and documents. The screens thus generated represent the degree of relationship (proximity, in graph theory terms) among activity sectors, companies, and the company leaders themselves.

Along with all of that and, more recently, access to non-awarding data (losing bids) opened the way to the use of techniques that made it possible to decisively promote pure *ex officio* detection. Eventually it led to the development and practical creation of BRAVA (Bid Rigging Algorithm for Vigilance in Antitrust).

BRAVA is a tool based on supervised machine learning that, through different models, can classify the different offers submitted to a tender as most probably collusive or competitive. The success rates obtained in the testing of these models are largely satisfactory, never less than 90%. It employs supervised learning based on raw variables (coming from the public procurement and registrars' databases already mentioned) and more than 20 screens coded, some based on international papers and academic articles and some others developed from scratch by the EIU considering its relationship with corruption / collusion.

After this process, data was divided into training and test sets, in order to proceed with the supervised machine learning process through different iterations. This process shows us the best performance variables from all the possibilities. In practice, this represents a double utility for determining collusion:

- It allows CNMC to target specific tenders, such as those included in a communication from procurement bodies or coming from whistle-blowers, and
- It enables to trigger pure *ex officio* analyses of certain sectors, especially on markets suspiciously under more stable cartels (whose members are less prone to defection).

Although BRAVA is always under continuous and intensive development, its first implementation is already in full operation, and its results are really promising.

Another investigative approach at the EIU which is quite different but takes advantage of some of the techniques mentioned above, is the use of Open-Source Intelligence (OSINT) and Human Intelligence (HUMINT) tools in order to provide an accurate identification and location of organizations and persons of interest, the relationships among them, and their degree of control of the companies and organizations which are under close scrutiny.

III. Main conclusions

The enforcement of competition law in digital markets is complex, especially as far as algorithms and artificial intelligence are concerned. Competition policy tools are flexible enough to adapt to the disruption driven by digitization, but some challenges still remain.

The CNMC has also given strategic priority to monitoring anticompetitive conduct driven by algorithms and digital business models, thus adapting and refining its enforcement tools. There is a real need for advanced computational tools based on similar techniques, to ensure a more even 'playing field' among cartelists and competition authorities, and therefore some of the current efforts of CNMC are set on that target.

In fact, the new, AI-based BRAVA system, recently set up by the CNMC and its Economic Intelligence Unit as described above, appears to be a definite effort in such a fight against bid rigging, one of the most relevant and widespread collusive activities.